

The Role of Disclosure in Decentralized Finance Markets: Evidence from Twitter

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ABSTRACT

Decentralized finance (DeFi) platforms provide lending, cryptocurrency exchange, and other services wherein all transaction details and underlying code are automatically posted on a public ledger (i.e., blockchain). Some DeFi platforms also voluntarily summarize and disclose blockchain transactions to help users understand blockchain information, which increase users' confidence in the platform. I document a positive relationship between a platform's blockchain-related posts on Twitter (now X) and its Total Value Locked (i.e., total cryptocurrency deposited in the platform). This relationship is stronger when tweets have greater information and when platform users face higher information processing costs. More tweets are also associated with increases in the number of users and voting participation. Furthermore, platforms strategically time tweets to manage liquidity pressures and strengthen their user base. These results suggest that blockchain information is costly to process and that platforms use strategic disclosure to aid users' information processing and coordinate their behavior.

JEL Classification: G14, M40, M41

Keywords: decentralized finance, cryptocurrency, voluntary disclosure, social media, processing costs

* For their guidance, support, and invaluable insights, I am very grateful to the members of my dissertation committee: Liandong Zhang (chair), Richard Crowley, Weikai Li and Holly Yang. I thank Anthony Le (discussant), Jongsub Lee (discussant), Daniel Rabetti (discussant), Anthony Zhang (discussant), Thomas Bourveau, Jonathan Chiu, Lian Fen Lee, Xuanbo Li, Pengkai Lin, Yun Lou, David Samuel, Katherine Schipper, Stefan Scharnowski, Donghwa Shin, Ke Wang, Rencheng Wang, Franco Wong, Christina Zhu, and the seminar participants at 2022 Cardiff Fintech Research Conference, 5th UWA Blockchain and Cryptocurrency Conference, 2022 Global AI Finance Research Conference, 2023 FARS Midyear Meeting, 2023 MIT Asia Conference, Singapore Management University and Hong Kong Baptist University for helpful comments. All errors are my own. Awyong is grateful for research support from Hong Kong Baptist University. There are no other conflicts of interest to disclose.

I. INTRODUCTION

The total market capitalization of cryptocurrency was US\$3.68 trillion in 2025 (Coinmarketcap 2025), surpassing the US\$3.13 trillion in assets under management of U.S. private equity firms in 2024 (Guevarra and Bharucha 2025). An important and rapidly growing segment within the crypto market is Decentralized Finance (DeFi), a group of blockchain-based applications providing services such as lending, exchange and asset management (Gogel 2021). These services are powered by code and executed on a public blockchain akin to a continuously updated and globally shared Google Sheet: anyone can inspect the data and add rows (i.e., transactions) by paying a small fee. Such technological advancement has led SEC Chairman Paul Atkins to state that “the American values of economic liberty ... and innovation are in the DNA of DeFi” and that DeFi stayed resilient during crises “while centralized platforms waivered and failed” (Atkins 2025a). These attributes have driven DeFi’s growth, with its Total Value Locked (TVL) increasing 125-fold over 5 years to reach US\$126.2 billion (DeFiLlama 2025). TVL represents assets deposited in the platform, and it plays a critical role in platform stability. Such exponential growth also has drawn regulatory scrutiny, particularly over the “lack of ... reporting requirements” (FSB 2023) that limit users’ ability to “assess risk likelihood and severity” when engaging with DeFi platforms (Crenshaw 2021), prompting calls for greater disclosure (Peirce 2025a). These concerns reveal a paradox: if users can obtain public blockchain data to estimate risks, then why is disclosure necessary?

In this paper, I examine whether and how DeFi platforms’ voluntary disclosure¹ of blockchain-related information on Twitter (now X)² affects their TVL. Higher TVL bolsters

¹ Some studies use “dissemination” specifically for the same information already disclosed elsewhere (e.g., Blankespoor, Miller, and White 2014), whereas others use “disclosure” even if the information disclosed elsewhere is similar (e.g., Christensen, Floyd, Liu, and Maffett 2017). I use a broad definition of disclosure that includes both summarized blockchain information (e.g., interest rate aggregated from transactions) and information found on the blockchain (e.g., transaction fees for a transaction).

² DeFi platforms also may use other social media (e.g., Telegram and Discord) to reach users. However, these channels are private and unobservable to researchers. Thus, my paper focuses on platforms’ use of public Twitter accounts to reach users.

liquidity by allowing users to easily withdraw or swap their crypto. As users become more financially committed through deposits and transactions, they also may be more inclined to participate in governance. Such participation mitigates governance risk by broadening the concentration of voting power beyond a few large users. Together, these mechanisms underscore the key feature of DeFi: platforms derive their fundamental value from a wide and heterogenous user base (Cong, Li, and Wang 2021).

On the one hand, although prior literature has documented that social media disclosure benefits investors, it is unclear whether the same result applies in the DeFi setting. Prior studies typically vary only one side of the dissemination process, either the supply of information by firms or the demand for information by users, whereas the DeFi setting jointly relaxes both sides. First, it relaxes the demand side because the underlying facts are already public and continuously observable on-chain, allowing users to directly extract data to estimate borrower default and liquidity risk without managerial intermediation (Bertomeu, Martin, and Sall, 2024). Second, the institutional absence of regulation and difficulty of verifiability also weakens both supply and demand. In traditional public firms, the ability to verify disclosure through SEC filings encourages firms to fully disclose (Grossman and Hart 1980; Milgrom 1981), lending credibility to their voluntary disclosure. In contrast, DeFi platforms face no mandatory reporting, so the analogous “verifiability” in the DeFi setting lies in the ability to accurately parse the blockchain, which may be impeded by the high cost of processing blockchain transactions. Finally, platform tweets are unaudited, which can lead to unconstrained managerial behavior on the supply side. Investors thus may discount disclosures as “cheap talk” and not rely on them. Overall, in unregulated environments where information is publicly available yet costly to process, voluntary disclosure may be dismissed as redundant at best and cheap talk at worst.

On the other hand, blockchain-related disclosure may increase TVL if it significantly lowers information processing costs. Even if underlying transaction-level data is publicly available, processing hexadecimal-encoded bytecode and Application Binary Interfaces is costly, and users may encounter inaccuracies if they rely on information intermediaries (discussed in Section II). As SEC Commissioner Crenshaw asserts, “It is not reasonable to build a financial system [demanding DeFi users] to be sophisticated interpreters of complex code” (Crenshaw 2021). Given these challenges, DeFi users have relied on Twitter for blockchain-related information, giving rise to an active community discussing crypto-related topics. In response to this demand, DeFi platforms have voluntarily summarized and disclosed blockchain information on Twitter, offering users an accessible way to obtain information.

To examine whether DeFi platforms’ voluntary disclosure of blockchain information affects their TVL, I collect financial information from DeFi Pulse and Coinmarketcap and collect platforms’ disclosures from Twitter. Specifically, I define my disclosure variable using only blockchain-related tweets, identified by applying a machine-learning categorization method (see Section III and Appendix A1). The disclosure variable includes blockchain-related topics such as liquidity, yield, gas fees, and governance and excludes non-blockchain topics such as advertising and community events (see Appendix A2 for the full list of topics). My final sample comprises 106 platforms and 2,117 platform-month observations.

I first test the association between the number of blockchain-related tweets and TVL. TVL is a measure of users’ confidence in DeFi platforms (George 2022). Thus, a positive association suggests that voluntary disclosure of blockchain information is helpful for DeFi users to assess risks such that users increase their confidence in, and financial commitment to, the platforms. Accordingly, I find that the number of blockchain-related tweets in the prior month is positively associated with TVL in the current month.

Next, I conduct two sets of cross-sectional tests to provide more insights into the main result. First, if blockchain-related disclosure is informative for DeFi users, then tweets with greater information content should be more highly associated with the platform's TVL. The more blockchain-related topics platforms discuss, the greater the information content of their tweets. The next textual characteristic follows the intuition in Dyer, Lang, and Stice-Lawrence (2017) that repeated sentences across documents can be redundant and have similar content. Consistent with expectations, results show that the subsamples with wider topic variety and greater content difference demonstrate significantly larger associations between the number of blockchain-related tweets and TVL. The second set of cross-sectional tests provides support for my argument that disclosure matters due to high information processing costs. Platforms' financial services are powered by "smart contracts," which users can query by writing scripts to retrieve and analyze blockchain information. As a proxy for the difficulty in querying the smart contract, I rely on the smart contract code, as well as the code that DeFi Pulse uses to calculate TVL. More lines of code correspond with higher processing costs that external parties incur when extracting smart contract information, as it suggests the contract structure is more complex and requires additional parsing steps. Overall, I find that my results are driven by the subsamples with more lines of TVL and contract code, supporting my argument that disclosure is helpful for users due to high costs of processing public blockchain data.

After establishing the main result that blockchain-related tweets increase TVL, I explore a potential mechanism explaining this effect. By providing accessible and aggregated information, blockchain-related tweets attract more users and increase their confidence to commit capital to the platform and to actively participate in governance. To test this, I first show that blockchain-related tweets increase both the number of new and existing DeFi users. I then show that the increase in TVL is likely driven by an extensive rather than intensive margin. Finally, I show that tweets increase the voting participation rate for proposals,

suggesting that improved information access encourages broader participation in platform governance. Taken together, the results suggest that blockchain-related tweets increase the value of DeFi platforms by expanding the user base and increasing user engagement.

Having shown that disclosure lowers information barriers and encourages user participation, I next examine whether founders use disclosure as a strategic device to coordinate decentralized users' behavior in the absence of centralized control and intermediation. As transactions in DeFi are automated, platforms cannot pause withdrawals or rely on designated market makers in times of stress. Therefore, liquidity depth and a stable user base are first-order concerns because they provide the primary defenses against liquidity shocks. First, if disclosure coordinates inflows under liquidity pressure, then platforms should tweet more (less) when large users have greater outflow (inflow). Using user wallet flow data unique to DeFi, I find exactly this pattern. Second, if disclosure is aimed at amassing a larger share of long-horizon users to stabilize the user base, then tweeting patterns should respond to user base composition. Consistent with this prediction, I find that blockchain-related tweets are positively associated with the percentage of mid-horizon wallets (those holding for 1 to 12 months) but negatively associated with the percentage of long-horizon wallets (those holding for more than 12 months). These results suggest that disclosure is used as a strategic tool to coordinate participation and stabilize liquidity in decentralized environments.

As an additional test, I examine whether the effect of disclosure is stronger when the need for transparency becomes more salient. The opaque nature of centralized exchanges (CEXes) suggests that hacks emphasize risks arising from users' inability to monitor their assets. Consistent with this, I find marginally significant results that post-CEX hack, platforms that disclose more blockchain-related information attract more users and increase their TVL.

To address endogeneity concerns, I first apply Oster's (2019) method and find a low likelihood that correlated omitted variables render my result non-significant. I next show that

my results hold using an instrumental variable approach, leveraging Twitter outages as a source of exogenous variation in disclosure. Finally, another concern is that tweets increase TVL due to an advertising effect instead of an information effect as I hypothesize (see Section V for details). However, cross-sectional results based on tweet content and lines of contract code are unlikely to reflect advertising motives.

My paper provides practical implications for the DeFi market at a time when regulators are ramping up efforts to safeguard the cryptocurrency ecosystem. Although the SEC has decided not to regulate DeFi (Peirce 2025b), it has asked for input on what types of disclosures are important for investor protection and what information to include in the ongoing disclosure (Peirce 2025c). In response, a group of key crypto stakeholders have proposed continuous disclosure of “source code, transaction history, and governance mechanisms” (Brennan 2025). My findings speak directly to this ongoing debate by providing empirical evidence that platforms summarize blockchain information (e.g., liquidity, yield, fees, and governance) and that such information increases TVL and the number of users. Aligning with the crypto stakeholders’ proposal, governance-related information appears to have the largest impact, though accessibility and aggregation also are important factors. I also find that platforms strategically time their disclosures to coincide with periods of large wallets’ outflow and high proportions of mid-horizon users, indicating that a clear disclosure framework may be necessary to limit platforms’ discretion. Together, these results can help inform SEC’s priority to build a “firm regulatory foundation for digital assets” (Atkins 2025b).

I next contribute to the emerging literature on blockchain technology and DeFi. This paper is the first to document how voluntary disclosure practices in the DeFi market affect its users. Similar research on initial coin offerings (ICOs) finds that greater disclosure in whitepapers (Bourveau, De George, Ellahie, and Macciocchi 2022; Howell, Niessner, and Yermack 2020) and social media (Benedetti and Kostovetsky 2021; Lyandres, Palazzo, and

Rabetti 2022) leads to better ICO performance. The DeFi setting fundamentally differs from the ICO setting in two ways. First, DeFi disclosure pertains to *public* blockchain information, whereas ICO disclosure likely pertains to *non-public* information. Second, the ICO setting focuses on *initial* fundraising, whereas the DeFi setting focuses on *ongoing* performance. As Bourveau et al. (2022, p.159) notes, “Although voluntary disclosure prior to the ICO reduces information asymmetry in the short term, it does not necessarily indicate an ongoing commitment to disclose.” We therefore lack evidence on how platforms use *ongoing* disclosure to meet goals of liquidity retention and user adoption. My paper fills this gap by providing the first evidence that ongoing information-rich tweets increase TVL and user adoption.

More broadly, my results add to the information processing literature. Prior research partitions investors by trade or holding size and finds that reduced costs disproportionately benefit smaller investors. (Crawford, Kim, Koo, and Le 2024), and social media analysis primarily benefits retail trades (Gomez, Heflin, Moon, and Warren 2024). In contrast, mandatory XBRL tagging lowers search costs for small buy-side firms (Bhattacharya, Cho, and Kim 2018). Yet, size is only one dimension of investor type. Ownership histories are observable only for 13-F filers. All other investors trade anonymously in the NYSE Trade and Quote dataset. Research has not yet addressed whether reducing processing costs attracts *new* investors or prompts *existing* investors to trade more. I fill this gap by identifying whether a wallet is a new or existing user and providing evidence that blockchain-related tweets are positively associated with both user types and have a non-significant relationship with TVL-per-user. These results, which imply that TVL growth is driven by an extensive rather than intensive margin, are important because they provide empirical support for Merton’s (1987, p.497)’s prediction that broadening the ownership base will increase firm value—a conclusion that cannot be directly tested using traditional stock market data.

Finally, prior work examining corporate use of social media has examined strategic incentives focusing largely on the nature and timing of public news. First, Madsen and Niessner (2019) find that customer-oriented tweets about upcoming sales launches and promotions increase sales growth. Second, Jung, Naughton, Tahoun, and Wang (2018) find that firms with negative earnings surprises tweet less, whereas Crowley, Huang, and Lu (2024) find that managers tweet more around significantly negative or positive news. In contrast, my paper offers new insights that in decentralized markets, where institutional infrastructure such as intermediated liquidity is absent, disclosure functions as a coordination device for decentralized users. These findings extend the strategic dissemination literature by showing that managers use social media not only to shape investor reaction to public news but also to coordinate participation to meet first-order needs such as liquidity depth and a stable user base.

II. INSTITUTIONAL BACKGROUND

Introduction to Blockchain and DeFi

DeFi is a category of decentralized financial service applications including lending and borrowing platforms. Users can deposit cryptocurrencies to earn interest or borrow against collateral (e.g., Aave), decentralized exchanges (DEXes) that facilitate cryptocurrency exchange (e.g., Uniswap), derivatives platforms (e.g., dYdX), asset management services (e.g., Yearn), and more. These services are provided through public blockchains. The most popular public blockchain, Ethereum, is like a public Google Sheet that any internet user can view. For a small “editing fee” (also known as a “gas fees” in DeFi), the user can add rows to this spreadsheet. For example, the Aave lending platform could occupy an individual worksheet containing three columns (e.g., Deposits, Borrowings, and Collateral) and a master cell (e.g., A1) that powers the whole system. Cell A1 contains the following interest rate formula: “IF(SUM(Borrowings) > 0.80 × SUM(Deposits), Base Rate + 0.50 basis points, Base Rate).” This means that a simple base rate is charged on all borrowings, along with an extra 0.50 b.p.

if the current utilization ratio exceeds 80 percent. Subsequently, any user submitting a transaction adds a new row that increases either Deposits or Borrowings, after which the worksheet automatically recalculates A1, and the updated interest rate applies to all future users. This Google Sheet is powered by back-end script (also known as smart contract code in DeFi), which governs the formulas and how rows are added. In the DeFi setting, the smart contract code is self-executing and directly enforces blockchain rules.

The rows added to the worksheet differ depending on the user's intended action. Deposits fill only the Deposits column, and borrowings fill the Borrowings column and name a cryptocurrency in the Collateral column. For example, if a borrower enters "USDC" in the Collateral column, the code executes a look-up to a reference table and confirms that USDC is acceptable collateral and that the requested loan amount satisfies the collateral ratio.³ If all conditions are met, the transaction is accepted and the loan is recorded as a new worksheet row. Continuing the analogy, platform founders are like the original creators of the Google Sheet. They pre-populate the master cell formulas and collateral table but later share the editing rights with its users via the issuance of governance tokens that function like voting shares. Each token grants its holder the right to propose and vote for/against edits to the workbook's master cells or reference tables. Continuing the Aave example, a successful vote can (1) change the 80 percent utilization threshold, (2) adjust the 0.5 b.p. parameter, or (3) change collateral rules like accepting a new cryptocurrency. Once these changes are accepted and reflected in the back-end script, all subsequently added rows must adhere to the new rules.⁴

³ To compensate for the lack of borrower screening, such as in traditional financial markets, many DeFi lending platforms require borrowers to over-collateralize to provide assurance to lenders. For example, Aave requires borrowers to over-collateralize at 116 percent of their loan amount when they use USDC as collateral (see <https://docs.aave.com/risk/asset-risk/risk-parameters>). Thus, to borrow 10 BTC, borrowers must lock in USDC of 11.6 BTC (~1.22mil USDC). Aave allows a slight buffer of 3 percent for BTC/USDC exchange rate fluctuations. If the exchange rate of BTC/USDC changes such that the value of 1.22 mil USDC is worth less than 113 percent of BTC, then the smart contract goes into an automatic liquidation process wherein anyone can act as a liquidator and repay the BTC loan on behalf of the borrower and earn a liquidation fee from the USDC collateral (see <https://docs.aave.com/faq/liquidations>).

⁴ Other DeFi services can be understood as similar worksheets. DEXes contain two columns, cryptocurrency X and cryptocurrency Y, and a master cell (e.g., A1) enforcing the exchange rate, $\text{Rate} = \text{Reserves X} / \text{Reserves Y}$.

Given the rapid growth of the DeFi market, a growing body of academic research has begun exploring its economic foundations. Theoretical research provides valuation models for understanding tokens, which enable users to conduct economic activities (e.g., voting, using collateral) on the platform. Cong et al. (2021) formalize this process with the term “tokenomics,” wherein the token price is determined by the platform’s productivity (e.g., technology and regulatory environment), total number of users, and user heterogeneity.

Another line of research delves deeper into users’ primary concerns about liquidity and governance risks. For example, Bertomeu et al. (2024) theoretically show that the risk of liquidation affecting both lenders and borrowers can be estimated using the sum of borrowings and deposits. Using transaction-level data from Maker and Aave, Qin, Zhou, Gamito, Jovanovic, and Gervais et al. (2021) show that sharp increases in the utilization ratio trigger large-scale liquidations that drain close to one-third of each platform’s TVL in less than 24 hours. In terms of governance risks, Sockin and Xiong (2023) theoretically highlight the importance of tokens as a commitment device that could prevent a platform from exploiting its users because tokens delegate control to users. However, Cong, Rabetti, Wang, and Yan et al. (2025) empirically show that voting is highly concentrated, with the top-decile voters controlling 76.2 percent of total voting power and a voting participation rate of only 6.3 percent.

A third strand of research examines the initial fundraising of ICOs, which is comparable to the IPO literature in traditional finance. Bourveau et al. (2022) show that lengthy and technical whitepapers increase capital raised, and Lyandres et al. (2022) find that social media activities also increase ICO success. Finally, a fourth line of research examines market signals such as smart contract code audits and finds that the market reacts positively to audit reports

Every new swap row updates the reserves and exchange rate simultaneously. An asset management platform operates a worksheet whose Deposit column automatically routes each new row to whichever external worksheet currently shows the highest header-cell yield, updating those routes at preset intervals. The common theme across these services is that each platform’s key logic—whether a borrowing-rate formula, exchange rate curve, or a yield optimiser—lives in one or two “master” cells, and user transactions simply append rows that reference these cells.

(Bourveau, Brendel, and Schoenfeld 2024). These audits typically verify smart contract code at a point in time and are infrequent technical certifications. In addition, crypto-influencers' tweets are initially associated with positive returns followed by significant negative longer-horizon returns (Merkley, Pacelli, Piorkowski, and Williams 2024).

Despite these advances, two important gaps remain. First, theory shows that a platform's long-term value is derived from maintaining liquidity and a heterogeneous user base. Empirical work thus far has focused almost exclusively on the *initial* fundraising moment (i.e., ICOs), showing that disclosures at this stage can improve ICO funding success. As Bourveau et al. (2022, p.159) note, "Although voluntary disclosure prior to the ICO reduces information asymmetry in the short term, it does not necessarily indicate an ongoing commitment to disclose." Similarly, smart contract audit reports are value-relevant but typically capture points in time and may not reflect sustained communication. Direct evidence on how platforms use *ongoing* disclosure to meet the goals of liquidity retention and user adoption is thus still lacking. My paper fills this gap by collecting blockchain-related tweets and on-chain user flows, providing the first evidence of how ongoing information-rich tweets can increase TVL and user adoption. Second, although prior research documents tightly concentrated voting power and low participation rates, it is unclear whether platforms take any mitigating measures to broaden participation. My paper fills this gap by showing that blockchain-related tweets (particularly governance tweets) can increase voting participation.

The interconnectedness of DeFi makes it susceptible to systemic risk. More empirical research is needed to understand this risk and how to mitigate it. For example, future research could assess the increasing reliance on cross-chain bridges (e.g., Wormhole, Multichain, Ronin) and the corresponding increase in hacks that cost billions to remediate.

Blockchain-Related Information and Disclosure

DeFi platforms are unregulated and thus not required to disclose any information. There is thus strong advocacy within the DeFi community to “do your own research” (DYOR) that comprises four tiers, arranged according to difficulty. The top tier is the full data dump, where highly technical users download the entire blockchain ledger, including machine-level structures like hexadecimal-encoded bytecode, low-level transaction call data, and application binary interface (ABI) specifications. Even simple questions like “How much did this wallet deposit?” require software engineering such as decoding hexadecimal logs, interpreting contract ABIs, and reconstructing transaction traces, which is nothing like standard coding in Python or R. The second tier consists of blockchain infrastructure providers (e.g., Alchemy) that serve as a gateway between the raw ledger and user applications (e.g., APIs). The fee subscription model reduces the burden of raw data ingestion but still requires users to code their own queries, resolve dependencies across addresses and smart contracts, and manage rate limits. A third and more accessible tier thus offers pre-built dashboards like DeFi Pulse or CoinMarketCap. These platforms offer visual summaries similar to pivot tables built from Excel worksheets, which allow users to explore TVL trends without writing any code. However, the summaries may reflect outdated smart contract logic or rely on broken token adapters, leading to incorrect conclusions. For example, the DeFi platform developer at YieldNest flagged an incorrect TVL calculation on DeFi Llama’s Github page,⁵ prompted by a 64 percent decrease in TVL reported on DeFi Llama’s webpage overnight. Furthermore, users at this tier are constrained by predefined metrics set by the data provider and cannot request additional information tailored to their needs.

Finally, the most accessible level of DYOR involves relying on social media posts that summarize blockchain information, and despite the lack of disclosure regulation, many

⁵ See <https://github.com/DefiLlama/DefiLlama-Adapters/issues/14936>.

platforms voluntarily share such information through the platform’s Twitter account. When users see the tweets — “Deposit rate raised ten b.p.” or “CAKE to be accepted as collateral” — they should immediately know that the master formula cell or collateral table has changed. These blockchain-related tweets can help users assess two key risks, liquidity and governance. First, users closely monitor platform liquidity due to crypto’s volatility and the need for fast withdrawals to seize trading opportunities. Real-time updates about deposit rates can help users decide whether to add liquidity and withdraw funds. Second, governance risk arises because large token holders can pass proposals without broader consensus in an unregulated market. To mitigate this risk, platforms often disclose active proposals and voting outcomes to promote transparency and encourage wider voting participation. In the absence of regulated disclosure, social media has emerged as the most accessible and timely way to convey information.

Hypothesis Development

This paper examines whether and how voluntary disclosure of blockchain information by DeFi platforms affects their TVL. Prior literature has documented that social media disclosure lowers information processing costs (Blankespoor et al., Miller, and White 2014; Jung et al. 2018). Yet, it is unclear if the same result applies in the DeFi setting because those studies typically vary either the supply of information by firms or the demand for information by users, whereas DeFi jointly relaxes both sides. First, it relaxes the demand side because the underlying facts are public and continuously observable on-chain such that users can parse the data to estimate borrower default and liquidity risk without needing managerial intermediation (Bertomeu et al. 2024). In this transparent environment, the marginal benefit of dissemination is uncertain: it could be redundant because users directly obtain the data they need, or it could be beneficial if directly interpreting on-chain data is complex and costly.

Second, in the DeFi setting, the institutional absence of regulation and difficulty of verifiability weaken both the supply and demand sides. For traditional public firms, voluntary

tweets are verifiable by audited SEC filings, which induces full disclosure by firms, a phenomenon dubbed “unravelling equilibrium” (Grossman and Hart 1980; Milgrom 1981). DeFi has no mandatory reporting, however, so this classical unravelling argument may not hold. Analogous verifiability in the DeFi setting is the ability to accurately parse the blockchain. Yet, the high cost of processing raw blockchain transactions or relying on information intermediaries that may provide incomplete information may result in users being unable to fully process blockchain information. The absence of regulatory oversight thus removes constraints on the supply side and erodes confidence on the demand side, as users cannot easily verify disclosures. Together, these features weaken the credibility of disclosures as compared to regulated markets. Another question that arises is that since DeFi is unregulated, why wouldn’t platforms selectively disclose private information only to large users. However, doing so undermines the trust among smaller users which would ultimately harm platform long-term value, which relies heavily on a broad user base. Because blockchain transactions by all wallet addresses are publicly visible, any attempts by large users to exploit private information would be immediately visible to the entire community.⁶

Third, platform tweets are unaudited, which could lead to unconstrained managerial behavior on the supply side. Founders may theoretically post any blockchain-related information they choose without formal oversight and in turn, investors may discount disclosure as “cheap talk” and not rely on blockchain-related tweets. In an unregulated environment where information is transparent but challenging to process, dissemination could be considered redundant at best or cheap talk at worst.

Nevertheless, tweets could still help users process blockchain information. Lu (2022) analytically shows that investors presented with various components of detailed information

⁶ As an extreme example, in 2022, a major lending platform Solend disclosed on Twitter that one user’s unusually large position risked crashing the entire network. The platform urged a governance vote to cap the borrowing limit, averting the crisis. This episode reveals how transparent on-chain transactions data enables public monitoring and fosters trust through public disclosure.

optimally allocate capacity across those components and do not fully process every piece. This results in them being better off with summary information (e.g., total cost of sales) rather than multiple components (e.g., line items adding up to cost of sales), even if the latter is more informative. Similarly, in the DeFi setting, blockchain-related tweets summarize disaggregated raw bytecode into digestible phrases (e.g., “Deposit rate raised ten b.p.”), which facilitate users’ understanding of the platforms and lead them to demonstrate greater confidence through higher financial commitments (e.g., increased TVL). Thus, I state my main hypothesis as follows:

H1: The number of blockchain-related tweets that DeFi platforms post on Twitter is positively associated with TVL.

III. RESEARCH DESIGN

Sample and Data

To construct my sample, I first obtain the names of all DeFi platforms listed on DeFi Pulse, a data aggregator for DeFi platforms.⁷ At the time of downloading, 131 DeFi platforms were listed. I then search for the platforms’ Twitter pages using links provided on DeFi Pulse supplemented with manual searches, identifying 121 platforms with active Twitter pages. After requiring non-missing platform control variables, I arrive at my main sample of 106 platforms providing a wide range of services (e.g., DEXes, derivatives, payments, asset management, and lending), with the sample period from May 2018 to March 2022. My final sample comprises a platform-month sample of 2,117 observations, as reported in Table 1 Panels A and B.

Key Variables

My main outcome variable is *TVL*, denominated in USD and provided by DeFi Pulse. I calculate the monthly average *TVL* for each platform and use the natural logarithm in my

⁷ DeFi Pulse stopped providing aggregated data shortly after I collected all my variables. It is now a crypto news outlet. Another popular data aggregator used by academic researchers, DeFi Llama, does not provide information on the number of wallet addresses, a key variable in my paper. Nevertheless, after re-estimating my main result using TVL collected from DeFi Llama, my results (untabulated) remain quantitatively similar.

main analysis. TVL is a stock measure and an important metric for DeFi platforms as it represents users' financial commitment to and confidence in a particular platform. When users lock their cryptocurrencies into the platform's smart contract, platforms have sufficient liquidity to maintain their operations. TVL is conceptually similar to the assets used by traditional firms to provide goods and services (e.g., the more deposits a bank has and the more machines a manufacturing firm has, the more goods and services they can provide). A healthy level of TVL also signals that the DeFi platform has sufficient liquidity and high collective confidence of its users.

I obtain from DeFi Pulse the monthly unique number of wallet addresses transacting with each platform as a proxy for the number of users ($\ln(\text{Unique Users})$). DeFi Pulse further decomposes unique users into new users ($\ln(\text{New Users})$), returning users ($\ln(\text{Returning Users})$), and resurrected users ($\ln(\text{Resurrected Users})$). Returning users are defined as wallet addresses that have interacted with the platform in the previous month, and resurrected users are those that have previously interacted with the platform but not within the previous month.

To measure voluntary disclosure, I first download all tweets posted by DeFi platforms within my sample period. As platforms also use Twitter for other purposes, such as hosting community events and customer support, it is important to restrict my analysis to tweets that contain blockchain-related information. To identify tweets with such information, I use an unsupervised machine learning technique, Twitter Latent Dirichlet Allocation (LDA), to categorize all tweets into 12 topics (see Appendix A1 for an overview of LDA and Appendix A2 for a list of topics). Next, I randomly select 200 tweets from each topic and assign a blockchain-related label if the content is related to the blockchain. For each topic, if the number of blockchain-related labels exceeds 80 percent,⁸ then all tweets belonging to that topic are

⁸ As a robustness check, I examine four other thresholds (i.e., 90 percent, 70 percent, 60 percent, 50 percent) and find the results remain quantitatively similar. I also train supervised text classifiers on the labelled sample, – a logistic regression (F1 = 86 percent) and a Sentence-BERT model (F1 = 93 percent) – and apply them to the full set of tweets. Using these predicted blockchain-related labels produces quantitatively similar results.

classified as blockchain-related.⁹ Appendix A3 presents sample tweets from each topic. My main disclosure variable is the natural logarithm of the total number of blockchain-related tweets ($\ln(\text{Blockchain Tweets})$).

I control for overall cryptocurrency market sentiment (*ETH Momentum*), which has been found to affect cryptocurrency returns (Liu, Tsyvinski, and Wu 2022). I also control for platform-specific factors that may affect both TVL and blockchain-related disclosure. First, audited platforms are deemed safer than are non-audited ones. Thus, to capture smart contract code robustness and control for audit status, I obtain from DeFi Pulse the dates when the platform's smart contracts were audited and construct a dummy variable that equals 1 for the post-audit period (*Audit*). Next, more established platforms may have higher TVL than newer platforms; thus, I control for the platform's age (*Age*) by taking its first tweet as its starting age.

I also include market-related controls from Coinmarketcap, the most common data source for cryptocurrency research (Lyandres et al. 2022). Market-related controls pertain to token characteristics. Most DeFi platforms issue a platform token that gives token holders the right to vote on governance issues. I calculate the token returns (*Token Returns*) to better isolate the effect of tweets on TVL, token price volatility (*Token Volatility*) to control for fluctuations in the platform's economic activities, market capitalization (*Market Cap*) to control for platform size, and token trading turnover (*Token Turnover*) to control for platform token liquidity. However, not all platforms issue tokens, so I also include a dummy (*Token Dummy*) for whether that platform has a token listed on Coinmarketcap and assign the median value for market-related controls if that observation has missing token-related data. I winsorize all continuous variables at the 1st and 99th percentiles. Appendix B lists all variable definitions.

⁹ Prior accounting literature examining earnings tweets by S&P1500 firms typically uses a keyword search to identify earnings tweets, but such approach is less feasible in the DeFi setting. For example, if I were to define blockchain-related tweets using the keyword "blockchain," as shown in Appendix A2, it likely will not capture any blockchain-related topics but instead capture the non-blockchain related topic, "Strategic Partnerships."

Research Design

To test H1 about whether voluntary blockchain-related disclosure affects the platform's TVL, I estimate the following regression for each platform i and month t :

$$\begin{aligned} \ln(TVL)_{i,t} = & \alpha + \beta_1 \ln(Blockchain\ Tweets)_{i,t-1} + \beta_2 Audit_{i,t-1} + \\ & \beta_3 ETH\ Momentum_{i,t-1} + \beta_4 Age_{i,t-1} + \beta_5 Token\ Dummy_{i,t-1} + \beta_6 Token\ Returns_{i,t} + \\ & \beta_7 Token\ Volatility_{i,t-1} + \beta_8 Market\ Cap_{i,t-1} + \beta_9 Token\ Turnover_{i,t-1} + \\ & Platform\ FE_i + Year\ FE_t + Month\ FE_t + \epsilon \end{aligned} \quad (1)$$

I use the lagged value of $\ln(Blockchain\ Tweets)$ in month $t - 1$ to mitigate reverse causality concerns. Accordingly, all control variables except for contemporaneous token returns are measured in the lagged period. In addition, I include platform fixed effects to control for time-invariant platform characteristics and year and month fixed effects to account for time-specific events affecting DeFi platforms. Standard errors are clustered at the platform level for all analyses. If disclosure of summarized blockchain information is not helpful for users, then I expect β_1 to be non-significant. However, if blockchain-related disclosure is useful and increases DeFi user confidence, then β_1 should be significantly positive.

IV. RESULTS

Descriptive Statistics

Table 1 Panel C illustrates the growth in DeFi platforms across my sample period. As shown, in 2018, there were only six DeFi platforms with a combined TVL of US\$3 billion and 49,000 users. This number steadily increased, from 69 platforms in 2020 to 106 in 2021. In the same period, their combined TVL grew from US\$16 billion to US\$79 billion and the number of users from 1.7 million to 3.6 million. Among the different DeFi platform categories, lending platforms consistently have the largest TVL and number of users, although the number of lending platforms does not dominate the sample. This result suggests that most DeFi activity comes from borrowing and lending transactions. Another category that has grown

exponentially is assets. These platforms boost their users' returns mostly through yield maximizing strategies, such as shifting cryptocurrency assets between lending platforms, similar to how traditional funds manage assets.

Table 2 Panel A reports the summary statistics for the main variables. DeFi platforms have an average TVL of US\$678 million across my sample period but are highly skewed, with a standard deviation of US\$2.2 billion. Thus, I use a log transformation of TVL to estimate my regression models. Of the monthly unique users, 64 percent are new users, 25 percent are returning users, and only 11 percent are resurrected users (untabulated). This suggests that DeFi users continue using platform's services or switch to another platform, but they do not often return to platforms that they previously used. DeFi platforms are also fairly active on Twitter during the sample period, posting an average 24 blockchain-related tweets per month, with a standard deviation of 29, and an average 35 words per tweet. Following Breuer and deHaan's (2024) recommendation to report the standard deviation within the fixed effect (FE) groups for blockchain-related tweets, I find the within-FE standard deviation is 18.04, or 62.4 percent of its raw variation, mitigating concerns that the variance in the explanatory variable of interest is excessively reduced (e.g., by 90 percent) by fixed effects (Breuer and deHaan 2024). The average age of DeFi platforms is 2.03 years, reflecting the relative immaturity of DeFi markets.

I further break down blockchain-related tweets into four topics and report the statistics in Table 2 Panel A. The most commonly tweeted topic is yield (8.23), followed by gas fees (5.24), liquidity (4.94), and governance (3.72). I report the pairwise correlation for my main variables in Table 2 Panel B, with correlations significant at the 1 percent level in bold. TVL is positively correlated with the lagged number of tweets and most lagged control variables. The number of blockchain-related tweets is positively correlated with contract code audit, age, token returns and volatility, market capitalization, and token turnover, suggesting that large platforms and platforms with greater fluctuations in economic activities tweet more.

H1: Impact of Voluntary Blockchain-Related Disclosure on TVL

I test H1 by estimating the impact of the number of blockchain-related tweets on TVL. If disclosure helps users process blockchain information and assess platform risk, then a significant association should occur between the two. Table 3 shows that the number of blockchain-related tweets in the previous period is positively associated with TVL both when I omit and include fixed effects (columns 1 and 2, respectively). Next, I include market-related controls of platform tokens in column 3 and find that the positive effect remains significant.¹⁰ Breuer and deHaan (2024) recommend that researchers use the within-FE standard deviation of variables when evaluating economic magnitude as it “avoids extrapolating the estimated effect above and beyond the variation in X that is actually used in the FE regression.” Accordingly, for a one-within-FE standard deviation increase in blockchain-related tweets, TVL increases by 28.1 percent.¹¹ In column 4, where I use an alternative measure of Twitter disclosure calculating the number of words used in blockchain-related tweets, the association with TVL is positive. Overall, the estimates should be interpreted as associations rather than causal effects since these results may be influenced by unobserved platform characteristics.

The association effects support H1 that platforms’ blockchain-related disclosure matters to DeFi users. Although blockchain-related tweets are derived from publicly available blockchain information and disclosures are unregulated, it appears that they are still informative for DeFi users. These tweets likely facilitate users’ assessment of platform credibility and thus increase their confidence in and financial commitment to these platforms. One question that may arise is why platforms do not disclose more to increase their TVL. A possible explanation is the immaturity of DeFi platforms, whose average age is two years.

¹⁰ I further decompose the blockchain tweet variable into its four topics: yield, gas fees, liquidity, and governance. All four are individually statistically significant when TVL is regressed on them, but only governance, liquidity, and yield remain significant when added to the same equation. In addition, only governance remains significant when number of users is regressed on it, suggesting that governance risk is a key consideration for DeFi users and that there may be fewer alternative sources for governance information.

¹¹ $e^{(0.4416 \times [\log(23.95 + 18.04) - \log(23.95)])} - 1 = 0.281$

These markets are only beginning to experiment with using Twitter as a disclosure channel and its effect on TVL. As platforms mature, their disclosure decisions may become more deliberate, thereby reducing the marginal effects in the later part of the sample. To empirically test this explanation, I re-estimate my baseline results for the earlier (2018–2020) and later (2021–2022) sample periods. The relationship between tweets and TVL remains significantly positive in both periods, but the coefficient is much larger in the early sample, with a one within-FE standard deviation increase in tweets corresponding to a 22 percent increase in TVL. In contrast, the corresponding impact declines to 9 percent in the later sample (untabulated), suggesting that DeFi markets are beginning to incorporate Twitter into their disclosure strategy.

How Information Content of Tweets Matters

After establishing that blockchain-related tweets can affect TVL, I explore how their information content strengthens this relationship. I use two variables to quantify information content. The first focuses on the number of Twitter-LDA topics that blockchain-related tweets cover (*No. of Topics*). Platforms post an average of 2.6 blockchain-related topics during the month. The more topics covered in the blockchain-related tweets, the greater the information content. I count the number of unique topics used by platforms within a month and partition my sample using the median *No. of Topics*. Columns 1 and 2 of Table 4 show that the relationship between blockchain-related tweets and TVL is significantly larger when platforms post about a greater variety of topics.

When analyzing firms' 10-K filings, Dyer et al. (2017) show that repeated sentences across documents are less informative for readers. By that same logic, tweets providing less repetitive and more varied content should be more informative for DeFi users. To capture the extent of dissimilarity in textual content, I use Universal Sentence Encoder, a machine learning technique developed by Google (Cer et al. (2018) and used in the accounting literature (e.g., Crowley et al. (2024)) to capture content differences between two sentences (see Appendix C

for an example). *Content Difference* represents the normalized Euclidean distance between all tweets posted by the platform within the same month using the Universal Sentence Encoder algorithm. I partition my sample based on *Content Difference*. As shown in columns 3 and 4 of Table 4, the relationship between tweets and TVL is significantly stronger when *Content Difference* is larger. Taken together, these results strengthen the main inference that voluntary disclosure is informative for users and thus increases platforms' TVL.

Are Results Driven by Information Processing Costs?

My main argument is that blockchain-related disclosure increases TVL due to the high costs of processing public blockchain information. To test this, I conduct two cross-sectional tests using proxies for information processing costs. If users intend to directly extract and process public blockchain information, they must write a computer script to query the platform's smart contract. As a proxy for this difficulty in querying the platform's smart contract, I rely on the code that DeFi Pulse uses to calculate each platform's TVL. This code focuses on extracting and calculating the platform's TVL, which can approximate the level of difficulty that DeFi users face when extracting other blockchain-related information from DeFi platforms. I thus partition my sample based on the lines of code that DeFi Pulse uses to calculate each platform's TVL (*Lines of TVL Code*). If the value of *Lines of TVL Code* exceeds the median, users are likely to incur higher processing costs. Consistent with expectations, the relationship between tweets and TVL is stronger for greater values of *Lines of TVL Code*, as shown in Table 5 columns 1 and 2, with a significant difference between subsamples. This result supports my argument that voluntary disclosure is helpful when it is more challenging for external parties to extract information from the platform's smart contracts.

Next, I obtain the smart contract code on which DeFi platforms operate. More lines of code in the platform's smart contract indicates higher processing costs for DeFi users and thus can approximate time spent on understanding and evaluating the platform's smart contracts. If

voluntary disclosure is informative for DeFi users due to high processing costs, a stronger relationship should be observed between blockchain-related tweets and TVL for platforms with more lines of contract code. Columns 3 and 4 of Table 5 show that the main results are indeed driven by the subsample of platforms with a higher-than-median value for *Lines of Contract Code*, and the difference between subsamples is significant. This finding supports my argument that users of these platforms face high information processing costs and thus increased reliance on platform disclosure. Both of these results support my argument that the association between blockchain-related tweets and TVL is due to information processing costs.

Tests Pertaining to DeFi User Base

Impact of Tweets on Number of Users

I next explore possible mechanisms for how tweets increase the value of DeFi platforms. Platforms derive their value from a wider and more heterogeneous user base. I therefore investigate whether improved access to blockchain information via tweets can attract and retain more users for DeFi platforms. DeFi users transact with platforms through their cryptocurrency wallets, each of which has a unique wallet address conceptually similar to a bank account number. I use the unique number of wallet addresses transacting with the platform's smart contract as a proxy for the number of users. This way, I can identify if a wallet address is interacting with the platform for the first time (i.e., new user), a consecutive recurring time (i.e., returning user), or a non-consecutive recurring time (i.e., resurrected user).

Columns 1 and 2 of Table 6 show that the number of blockchain-related tweets in the previous month is positively associated with the number of unique platform users both when I omit and include fixed effects. Furthermore, columns 3 to 5 present the breakdown by user type and show that blockchain-related tweets lead to increases in both new and returning users. This finding suggests that blockchain information disclosed on Twitter is helpful for both potential platform users and existing users who may face high costs in continually processing blockchain

information and thus rely on summarized blockchain-related disclosure. In contrast, blockchain-related disclosure does not have a significant impact on resurrected users, perhaps because these users return to the platform for reasons other than information processing.

User Base Concentration and Voting Participation

Platform value increases when the user base is wider. Therefore, I explore whether the increase in TVL is driven by an intensive (more concentrated user base) or an extensive (wider user base) margin. Three possible scenarios can explain an increase in TVL: more new users (i.e., extensive), retaining existing users (i.e., extensive), and both new and existing users depositing TVL (i.e., intensive). If disclosure helps users process blockchain-related information, then ex-ante disclosure should increase TVL by enabling more users to understand and participate in the platform (i.e., extensive growth).

I first partition the sample based on the TVL concentration per user (*TVL Concentration*), calculated by scaling platform's TVL by the number of unique platform users. Columns 1 and 2 of Table 7 Panel A present this subsample analysis and show that the main results are driven by low TVL concentration per user, with a statistically significant difference between subsamples. This result helps rule out the explanation that TVL increases due to an intensive margin. Next, I directly regress the number of blockchain-related tweets on *TVL Concentration* and find a non-significant association, as shown in column 3. Overall, the results in Table 7 Panel A suggest that TVL growth is less likely driven by new and existing users investing more in the platform and more likely driven by attracting a greater number of new users and retaining a greater number of existing users.

Next, I test if tweets increase the rate of voting on governance proposals. Governance risk is important to DeFi users due to a lack of regulation. DeFi users and platform founders have an incentive to encourage greater user participation to lower governance risk. If governance-related tweets can facilitate users' understanding of governance proposals, it may

increase their confidence in voting on these proposals. I thus test whether blockchain-related tweets are associated with higher voting participation rates (*VotePart*). The number of votes cast is downloaded from DeFi Llama and then divided by total circulating supply of tokens to obtain *VotePart*.¹² Voting data can be matched only to a small subset of my sample, however, so the results should be interpreted with caution. Table 7 Panel B column 1 shows that all blockchain-related tweets indeed increase the voting participation rate. In column 2, using only governance-related tweets, I repeat the test and find a similar positive association. The results suggest that when platforms tweet about governance-related topics, more token holders are encouraged to exercise their voting rights. Overall, the results suggest that disclosure encourages voting participation and lowers governance risk.

Strategic Tweeting

DeFi platforms face two first-order concerns: maintaining liquidity and strengthening the user base. Real-time liquidity pressure arises when outflows from large users or liquidity providers quickly drain platform funds, thin liquidity depth, and increase the risk of forced liquidations.¹³ Because transactions are automatically executed by smart contract code, these trades clear even under liquidity stress as no manager or central bank can intervene by slowing withdrawals or pausing redemptions. Without an analogous “lender of last resort,” platforms must rely on sufficient liquidity to absorb shocks and preserve price discovery (Cong and He 2019), with liquidity preventing forced liquidations and stemming rapid declines in TVL.

A stable long-term user base anchors liquidity depth and reduces exit risk. Committed users supply transactional demand and are more likely to provide liquidity, both of which increases platform valuation (Cong et al. 2021). At the same time, user beliefs and network

¹² Doing so assumes that each token carries one vote, but there could be cases where each token carries more than one vote. To address this issue, I exclude proposals with participation rates exceeding 100 percent.

¹³ A forced liquidation occurs when the collateral value falls below the required threshold. The smart contract then automatically sells the collateral to repay the debt. This practice protects the lenders but can amplify stress by pushing prices lower and triggering additional liquidations.

effects shape participation (Sockin and Xiong 2023). Platforms wanting to retain marginal users who are deciding whether to exit or stay can use disclosure to reach those users. This proposed mechanism motivates testing whether platforms strategically increase disclosure when liquidity outflows intensify or when there is a greater presence of marginal users so that they can coordinate decentralized user behavior to address key operational challenges.

I first examine if real-time liquidity pressures affect platforms' disclosure decisions. As ownership is highly concentrated in DeFi platforms (Cong, Tang, Wang, and Zhao 2022), any withdrawals by large wallets may threaten a platform's stability and compel it to increase disclosure to attract inflows. I obtain wallets flow data from Intotheblock, where large wallets are those holding at least 0.1 percent of total supply. Next, I define *Large Wallets Outflow (Inflow)* as total outflows (inflows) from large wallets scaled by total supply. I then regress the number of blockchain-related tweets on large wallets' flow. Table 8 column 1 shows that platforms tweet more when there is greater outflow from large wallets, and column 2 shows that platforms tweet less when inflows rise. This suggests that platforms strategically increase disclosure to attract inflows when liquidity is threatened but disclose less once it stabilizes.

Next, I examine if platform founders strategically tweet to strengthen their user base. I focus on wallets with mid-horizon holding periods, defined by Intotheblock as those retaining tokens between 1 to 12 months, because they represent marginal users still evaluating their long-term commitment. In contrast, long-horizon wallets (holding for over 12 months) are indicative of a committed user base that founders are likely seeking to cultivate, and short-horizon wallets (holding less than 1 month) likely represent frequent traders. Accordingly, I regress the number of blockchain-related tweets on the percentage of mid-horizon wallets. Table 8 column 3 shows a positive association, whereas column 4 shows that the percentage of long-horizon wallets are negatively associated with tweets. Both results are derived from a small sample, marginally significant at the 10 percent level, and thus should be interpreted with

caution. Taken together, the results provide some evidence that founders tweet to reduce information processing costs for mid-horizon wallets, encouraging them to remain invested and transition into long-horizon wallets. Once this objective is met and the percentage of long-horizon wallets increases, founders have less incentive to tweet.

V. ADDITIONAL TESTS AND ROBUSTNESS CHECKS

Contagion Risk from Centralized Cryptocurrency Exchanges Hacks

One potential downside of an interconnected DeFi network is high contagion risk. For example, the collapse of FTX (a centralized cryptocurrency exchange or CEX) threatened the confidence of the entire cryptocurrency ecosystem such that DeFi's overall TVL also suffered. More generally, CEXes are opaque. Their transactions are not visible on the public blockchain. Thus, during adverse events such as cybersecurity hacks, CEXes may covertly grant preferential trading access to certain users while halting trades on other users' assets.¹⁴ This opacity disallows public monitoring precisely when users are most concerned about the safety of their cryptocurrency assets.

I explore whether this change in salience in opacity risk results in users choosing DeFi platforms that disclose more. Both existing DeFi users and centralized crypto users may switch to DeFi platforms that tweet more blockchain information to better understand and monitor the platforms. I test this by creating a dummy variable that equals 1 if any CEX suffered a cybersecurity hack in the previous month (*CEX Hack*), then interact this variable with the main disclosure variable ($\ln(\text{Blockchain Tweets})$). I include DeFi category (similar to industry) fixed effects to examine whether DeFi users choose more transparent platforms within each category that provides similar services. Table 9 column 1 shows a positive and marginally significant

¹⁴ In contrast, when DeFi platforms are hacked, users have direct access to subsequent blockchain transactions and can analyse how the hack occurred. For example, when \$610 million was stolen from the DeFi platform Poly Network, its users scrutinized the platform's contract code and determined the attack was due to mismanagement of access rights to key smart contracts.

coefficient on the interaction term, suggesting that transparent platforms can attract more users, particularly returning and new ones (columns 2 and 3). Similarly, the interaction term between hacks and disclosure has a marginally positive impact on TVL, as seen in Table 9 column 4. Overall, I find some evidence that cryptocurrency users place a stronger emphasis on platform disclosure when there is an increase in salience of risks arising from opacity.

Robustness Checks

My inference that platforms' blockchain-related tweets increase TVL is potentially confounded by endogeneity concerns. I thus follow the recommendations in Armstrong, Kepler, Samuels, and Taylor (2022) to exploit the features of my setting and triangulate my results across different research designs. First, there is a possibility for reverse causality if TVL drives the number of blockchain-related tweets. Periods with high TVL may signal greater user interest in DeFi platforms, which in turn elicits higher demand for information. This concern is more severe if I include advertising tweets in my independent variable as platforms may try to capitalize on high cryptocurrency sentiment to attract more users. However, my main blockchain-related tweets variable excludes pure advertising tweets and is more likely to contain only blockchain-related information. Second, the presence of unobservable omitted correlated variables could affect both TVL and the number of blockchain-related tweets. These variables could pertain to macro-level factors, such as a cryptocurrency bull market or platform characteristics (e.g., size of founding team). Thus, the positive association that I document between TVL and tweets may be explained by these correlated omitted factors, instead of a true relationship between the two. I mitigate these endogeneity concerns in several ways.

Tests Mitigating Correlated Omitted Variables Concerns

First, I repeat my main results using a platform-day level sample and use the change in TVL regressed on the change in blockchain-related tweets (untabulated). These two approaches are less likely to be subject to omitted variables than approaches using a longer time interval.

Second, I conduct cross-sectional tests to show that my results vary with factors in line with my main argument. The relationship between blockchain-related tweets and TVL is stronger when tweets have greater information content and when platforms have more complex code. Any correlated omitted variable must also be correlated with these two factors.

Next, I examine the potential effect of omitted variable bias by constructing bias-adjusted coefficients following Oster (2019). This bias adjustment is derived by comparing the coefficient on $\ln(\text{Blockchain Tweets})$ and the resulting R^2 of a regression excluding control variables (Table 3 column 2) versus a regression containing measurable control variables (Table 3 column 3). Following Oster (2019), I set $R_{\max}$ equal to 1.3 times the R^2 of the regression including control variables, and 1.5 times the R^2 in a more conservative approach.

Table 10 Panel A shows deltas of 12.998 and 8.035 when $R_{\max}$ is set to 1.3 and 1.5 times of R^2 , respectively. The results suggest that the unobservable factors would need to be at least 8 times as important as the observables to render $\ln(\text{Blockchain Tweets})$'s impact non-significant. The magnitude of these deltas thus provides reassurance that the effect of blockchain-related tweets is unlikely to be driven by omitted variables.

Instrumental Variable Approach

Notwithstanding the low likelihood that an unobservable factor renders my main result non-significant, I adopt an instrumental variable (IV) approach to capture exogenous variation of platforms' Twitter activity due to Twitter outages. Such operational disruptions are likely to reduce DeFi platforms' Twitter activity but are unrelated to the platforms' TVL. Following Jia, Redigolo, Shu, and Zhao (2020), who similarly use Twitter outages as an IV, I collect Twitter platform interruptions from Twitter's official support account (@TwitterSupport), as it informs users about issues on Twitter. I then manually review all tweets and identify those describing outages. As Twitter outages likely affect DeFi platforms only momentarily, I use platform-day level data for the IV analysis (as compared to platform-month level for my main results).

In the first column of Table 10 Panel B, I report the first-stage regression using the instrument *Twitter Outage*. The coefficient on the instrument is negative and significant at the 1 percent level, and the partial F statistic on the excluded instrument is 10.06, which satisfies the conventional criteria for instrument relevance. This suggests that DeFi platforms post fewer blockchain-related tweets during Twitter outages. Column 2 presents the second-stage estimation, where I regress daily TVL on the instrumented blockchain-related tweets from the first stage. The coefficient remains positive and significant at the 5 percent level, mitigating concerns about reverse causality and correlated omitted variables.

Alternative Explanation: Advertising Effect

Another explanation for why blockchain-related tweets increase TVL is that tweets have an advertising effect. Prior literature has shown that firms strategically use advertising to influence stock prices (Madsen and Niessner 2019). Two types of advertising tweets exist in my setting: purely promotional (e.g., “Kyber Network is currently the most popular exchange to trade DAI on!”), which are non-informative and excluded from my sample, and blockchain-related (e.g., “Bancor has now an \$SNX liquidity pool!”), which contain potentially useful blockchain-related information and are included in my sample. Cross-sectional results show that tweets with greater information content and platforms with more complex contract code have stronger associations with TVL and support the information explanation, as these variables have little apparent relation to advertising.

I further address this concern in two ways. First, if tweets serve an advertising purpose, then bullish tweets should primarily drive TVL. However, if tweets serve an informational role, then neutral or bearish tweets should have a greater impact. I follow Merkley et al. (2024) and use Crypto-BERT to categorize each blockchain-related tweet as bullish, neutral, or bearish. I find that the impact of neutral tweets on TVL is double that of bullish tweets (untabulated). This suggests that DeFi users value tweets for their informational content, consistent with an

information rather than advertising explanation. Second, my finding that tweets increase governance voting participation (see Table 7 Panel B) is difficult to reconcile with a pure advertising effect, as voting participation does not directly follow from promotional attention. Instead, it is more consistent with an informational role (i.e., tweets summarize, interpret, and disseminate governance information to reduce frictions that prevent users from voting).

VI. CONCLUSION

DeFi platforms are built on transparency and thus provide a unique research setting. Full transaction details and underlying smart contract code are publicly observable from the blockchain in real-time. Despite such unprecedented transparency, however, regulators worry about the lack of disclosure requirements, particularly its impact on small DeFi users. This apparent puzzle in requiring transparent platforms to disclose public information can be explained by information processing costs. Blockchain data carries high information processing costs to extract useful insights, and users relying on information intermediaries may not obtain accurate or relevant information. Thus, in the absence of regulatory disclosure, blockchain-related tweets posted by the platforms offer an accessible and reliable way for DeFi users to obtain blockchain information, which in turn helps create value for the platform.

I find that voluntary disclosure of blockchain information on Twitter is positively associated with the platform's TVL. These results are robust to using instrumented blockchain-related tweets. The relationship between blockchain-related tweets and TVL is stronger when the tweets provide greater information content and when users face higher information processing costs. Furthermore, I find that tweets are positively related to both new and existing users and to users' platform-related voting participation. DeFi platforms also tweet more when outflows by large wallets rise, and they tweet less when large wallets deposit more. Tweet volume increases with a higher share of mid-horizon wallets, and it decreases when long-horizon wallets dominate. These results suggest that Twitter could be used as a coordination

tool to manage liquidity pressures and to strengthen the user base, which is crucial while operating in decentralized environments.

Overall, this paper provides the first empirical study of DeFi's information environment and finds that blockchain-related tweets help users understand complex blockchain information. Such voluntary disclosure appears to fill an information gap, a major concern of regulators. The results thus provide empirical backing at a time when the SEC is focused on designing effective ongoing disclosure for DeFi platforms.

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APPENDIX A1: TWITTER – LATENT DIRICHLET ALLOCATION METHOD

To categorize the content of platform-issued tweets, I employ the Twitter-LDA algorithm of Zhao et al. (2011), which extends the LDA framework of Blei, Ng, and Jordan (2003) to short-text data. Unlike dictionary-based approaches that impose researcher-defined word lists, Twitter-LDA identifies topics in an unsupervised Bayesian manner by examining how frequently words within the same tweet co-vary with one another. Furthermore, Crowley et al. (2024) applied Twitter-LDA to categorize millions of public firms' tweets and found that it classifies much more accurately than the dictionary approach.

In my paper, I first determine that the optimal number of topics for Twitter-LDA to find is 15. The optimal number is determined through a combination of coherence and perplexity scores from model topics ranging from 5 to 30. I then group conceptually related topics based on their top keywords and representative tweets, resulting in a final set of 12 topics, which are listed in Appendix A2. See Appendix A3 for sample tweets for each Twitter-LDA topic.

APPENDIX A2: KEYWORDS OF LDA TOPICS

S/N	Topic Label	Top 5 Keywords	% of Sample	Blockchain Related?
1	<i>Yield</i>	yield; deposit; price; supply; apy	13%	Yes
2	<i>Gas Fees</i>	liquidity; trade; gas; fee; price	9%	Yes
3	<i>Liquidity</i>	liquidity; deposit; token; pool; stake	8%	Yes
4	<i>Governance</i>	vote; proposal; governance; token; stake	6%	Yes
Total Blockchain-Related			36%	
5	<i>Platform Founders</i>	community; team; work; defi; build	13%	No
6	<i>Strategic Partnerships</i>	defi; crypto; index; blockchain; asset	10%	No
7	<i>Community Events</i>	join; community; ama; discord; defi	9%	No
8	<i>Customer Support</i>	index; token; support; trade; issue	9%	No
9	<i>Trading Opportunities</i>	alpha; farm; reward; pool; stake	6%	No
10	<i>Platform Awareness</i>	liquidity; trade; reward; farm; earn	6%	No
11	<i>Contests</i>	nft; win; winner; trade; prize	6%	No
12	<i>Cryptocurrency Advertisements</i>	app; card; bitcoin; btc; crypto	5%	No
Total Non-Blockchain Related			64%	

APPENDIX A3: SAMPLE TWEETS FOR THE 12 TOPICS

S/N	Topic Label	Sample Tweets
1	<i>Yield</i>	“Weekly Performance of ETH Theta Vault: The vault sold the \$2.2k Strike Call and ETH expired at ~\$2.06k, so the options expired out of the money. The vault earned ~0.37% (~21% APY) of yield in ETH. Another great week for stacking ETH!”
3	<i>Liquidity</i>	“Total supply on #LendfMe, the money market of @dForcenet, hits a new all-time high of \$13M. Earn more #crypto by saving it.”
3	<i>Governance</i>	“BIP#1 has passed with almost 40% of token holders voting in favor. As of block 12533751, BPRO will be distributed to users who provide BPRO to Uniswap V2 and Sushiswap ETH-BPRO pools. 41k BPRO will be distributed monthly. Currently every supplied BPRO gets 1.32 BPRO monthly.”
4	<i>Gas Fees</i>	“@0xScott @UniswapExchange Gas price shouldn’t be higher than 0.1 gwei”
5	<i>Platform Founders</i>	“b'RT @avichal: good example of why vesting for founding teams matters. Also at using Litecoin as an example of a project that survived.”
6	<i>Strategic Partnerships</i>	“Sharpe Announces a Cooperation Partnership with ETHLend!! #Sharpe #ETHLend #blockchain”
7	<i>Community Events</i>	“Join #Alpha AMA today (Mar 17th) at 3pm UTC on Alpha Finance Lab discord.”
8	<i>Customer Support</i>	“@_MrHand_ Working well here. DM us if the issue persists.”
9	<i>Trading Opportunities</i>	“Alchemix lets you manifest your future yield into the present. We've been hearing from some community members about what they plan to finance with our self-repaying loans. What will you do with your \$aUSD?”
10	<i>Platform Awareness</i>	“@duniacryptoid @FortuneTube @TraderTrek @anrubudda @Gomblo112 Thank you for the attention to ForTube, if you have any question you can contact ForTube telegram: https://t.co/UqM3RADyqA . ”
11	<i>Contests</i>	The end date of the #1inch #Discord #Meme #Contest moves to July 30! Got no potential winners so far. Cheaters shall not pass! Honest meme makers still have good chances to win.
12	<i>Cryptocurrency Advertisements</i>	“Every informed person needs to know about #Bitcoin because it might be one of the world’s most important developments by #LeonLuow. What’s your guess on this? We guess that he is right, Bitcoin and #Blockchaintechnology are one of the biggest developments.”

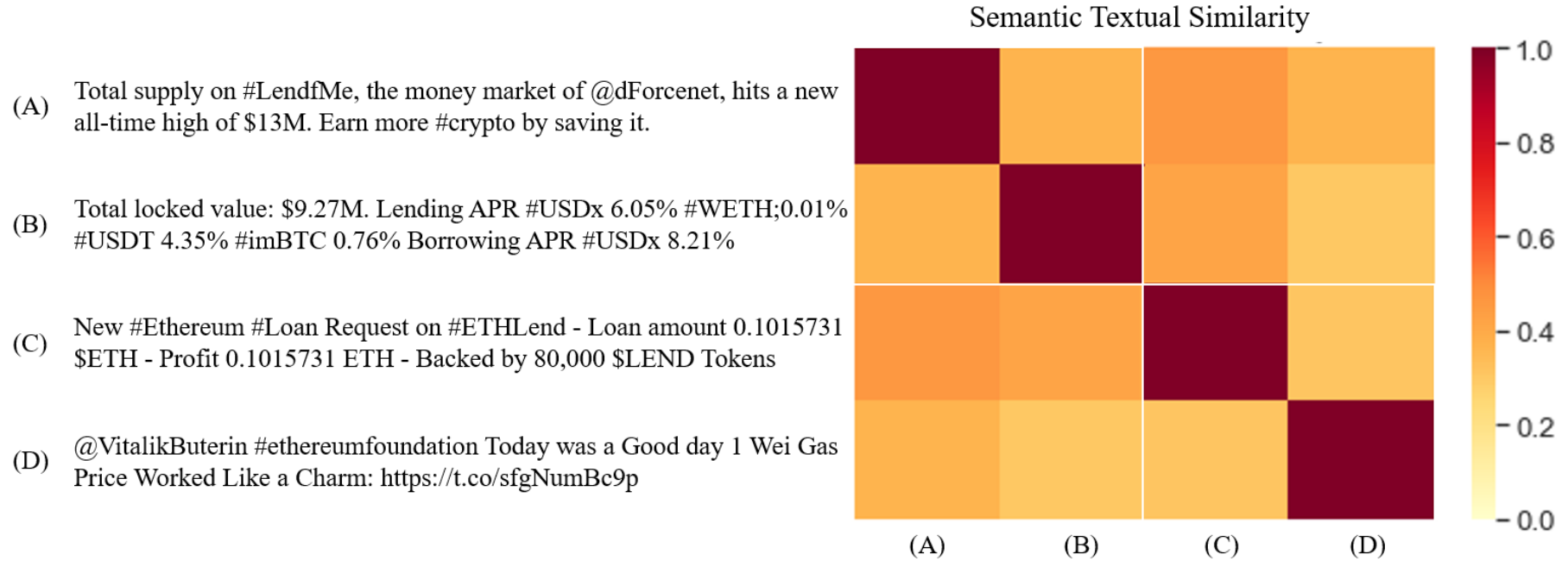
APPENDIX B: VARIABLE DEFINITIONS

Variable	Description
<i>Dependent Variables</i>	
$Ln(TVL)$	The natural logarithm of total value locked (TVL) within the DeFi platform, denoted in USD, and averaged over the month. DeFi Pulse calculates TVL by extracting the total balance of crypto assets held by the DeFi platform's smart contracts.
$Ln(Unique\ Users)$	The natural logarithm of the unique number of wallet addresses that interacted with the platform's smart contract within the month. Unique users is the sum of (1) new users, (2) returning users, and (3) resurrected users, as defined below.
$Ln(New\ Users)$	The natural logarithm of the unique number of new wallet addresses that interacted with the platform's smart contract for the first time.
$Ln(Returning\ Users)$	The natural logarithm of the unique number of returning wallet addresses that interacted with the platform's smart contract in both the last month and current month.
$Ln(Resurrected\ Users)$	The natural logarithm of the unique number of wallet addresses that interacted with the platform's smart contract more than a month ago that stopped interacting last month and then resumed interacting in the current month.
$TVL\ Concentration$	TVL scaled by total number of unique users.
$VotePart$	The total number of votes cast for each governance proposal divided by total circulating supply of tokens.
<i>Independent Variables of Interest</i>	
$Ln(Blockchain\ Tweets)$	The natural logarithm of 1 plus the total number of blockchain-related tweets that the DeFi platform has posted in the month.

<i>Ln(Total Words)</i>	The natural logarithm of the total number of words used in <i>Ln(Blockchain Tweets)</i> .
<i>Ln(Governance Tweets)</i>	The natural logarithm of 1 plus the total number of governance-related tweets that the DeFi platform has posted in the month.
<i>Large Wallets Outflow</i>	Total outflows from large wallets scaled by total supply of tokens. Large wallets are those holding at least 0.1% of the total supply of tokens.
<i>Large Wallets Inflow</i>	Total inflows from large wallets scaled by total supply of tokens.
<i>% Mid-Horizon Wallets</i>	Total number of mid-horizon wallets scaled by the total number of wallets. Mid-horizon wallets are those holding tokens for between 1 to 12 months.
<i>% Long-Horizon Wallets</i>	Total number of long-horizon wallets scaled by total number of wallets. Long-horizon wallets are those holding tokens for more than 12 months.
<i>Control Variables</i>	
<i>Audit</i>	Dummy that equals 1 after a smart contract code audit has been conducted for the DeFi platform.
<i>ETH Momentum</i>	Ether momentum using ETH returns compounded over the month.
<i>Age</i>	Platform age defined as the number of years since the platform's first inception on Twitter (now X).

<i>Token Dummy</i>	Dummy that equals 1 if the DeFi platform has a platform token with price information available on Coinmarketcap, and 0 otherwise.
<i>Token Returns</i>	The compounded platform token's returns in that month.
<i>Token Volatility</i>	Annualized standard deviation of platform token's returns over the month.
<i>Market Cap</i>	The number of platform tokens in circulation multiplied by the closing price of the platform token as at the end of the month (scaled in millions).
<i>Token Turnover</i>	The total trading volume of the platform token during the month divided by the number of platform tokens in circulation (scaled in thousands).
<i>Cross-sectional Variables</i>	
<i>No. of Topics</i>	The number of blockchain-related topics (as defined by Twitter-LDA) that the platform has tweeted about during the month.
<i>Content Difference</i>	The normalized Euclidean distance between all tweets posted by the platform within the same month using the Universal Sentence Encoder algorithm. The more textual content differs, the larger <i>Content Difference</i> is.
<i>Lines of TVL Code</i>	The number of lines of code that DeFi Pulse uses to query and extract the TVL from each DeFi platform.
<i>Lines of Contract Code</i>	The number of lines of code the platform uses in its smart contract.
<i>Additional Test Variables</i>	
<i>CEX Hack</i>	Dummy that equals 1 if a centralized cryptocurrency exchange experiences a cybersecurity hack.
<i>Twitter Outage</i>	Dummy that equals 1 if Twitter experiences a service outage.

APPENDIX C: EXAMPLE OF UNIVERSAL SENTENCE ENCODER ALGORITHM



In the example above, (A) and (B) are posted by a DeFi platform within the same month, and (C) and (D) are posted by another platform in another month. The Universal Sentence Encoder algorithm (Cer et al. 2018) correctly identifies that (A) and (B) are more similar in content than (C) and (D) are, indicated by darker shades of orange. (A) and (B) disclose the financial performance of the platform, with (A) describing the total market supply and (B) disclosing various financial metrics. In contrast, (C) highlights a specific blockchain transaction, and (D) discusses transaction costs. Thus, (C) and (D) provide greater information content than (A) and (B) do. Interestingly, the algorithm also identifies that (C) provides similar content to (A) in that they both suggest yield opportunities, even though the two tweets contain no overlapping words besides “on,” “new,” and “by.”

TABLE 1**Sample Selection and Distributions**

This table presents the sample selection procedure (Panel A), annual distribution (Panel B), and distribution by DeFi category and year (Panel C). The final sample includes 2,117 platform-month observations between 2018 and 2022.

Panel A: Sample Selection

Year	No. of Platform-Month Observations	No. of Unique Platforms
Platform-months in DeFi Pulse between Jan 2018 and Mar 2022	2,656	131
Less:		
Missing Twitter data	(339)	(10)
Missing control variables	(200)	(15)
Total	2,117	106

Panel B: Sample Distribution by Year

Year	No. of Platform-Month Observations	Percentage (%)	Cumulative Percentage (%)
2018	38	1.79	1.79
2019	144	6.80	8.60
2020	464	21.92	30.51
2021	1,166	55.08	85.59
Q1 2022	305	14.41	100
TOTAL	2,117	100	-

TABLE 1 – Continued

Panel C: DeFi Platform Statistics Distribution by Year and Category

Year	DeFi Platform Category	Total Unique No. of Platforms	Average TVL (USD billions)	Total Unique No. of Users (in Millions)
2018	Lending	3	2.982	0.019
	DEXes	1	0.025	0.027
	Assets	1	0.007	0.002
	Derivatives	1	0.002	0.002
	TOTAL	6	3.016	0.049
2019	Lending	5	7.744	0.166
	DEXes	2	0.009	0.034
	Assets	3	0.007	0.001
	Derivatives	4	0.061	0.008
	Payments	4	0.002	0.001
	TOTAL	18	7.823	0.210
2020	Lending	14	13.138	1.286
	DEXes	15	1.398	0.228
	Assets	23	1.098	0.102
	Derivatives	13	0.564	0.136
	Payments	4	0.093	0.008

		TOTAL	69	16.293	1.759
2021	Lending	28	45.050	1.906	
	DEXes	20	16.337	0.658	
	Assets	31	13.442	0.516	
	Derivatives	23	2.760	0.425	
	Payments	4	2.124	0.097	
		TOTAL	106	79.714	3.602
Q1 2022	Lending	27	40.133	0.244	
	DEXes	18	14.479	0.097	
	Assets	31	17.749	0.192	
	Derivatives	22	2.156	0.038	
	Payments	4	1.670	0.015	
		TOTAL	102	76.188	0.585

TABLE 2

Summary Statistics and Correlations

This table reports the summary statistics (Panel A) and pairwise correlations (Panel B) of the main variables used in the analyses. Correlations significant at the 1 percent level are presented in bold font. Please refer to Appendix B for variable definitions.

Panel A: Summary Statistics

	N	Mean	St. Dev	p25	Median	p75
Main Variables						
<i>TVL (USD)_t</i>	2,117	677,637,438	2,179,975,884	3,175,596	25,055,509	188,944,982
<i>Blockchain Tweets_{t-1}</i>	2,117	23.95	28.92	4.00	13.00	33.00
<i>Total Words_{t-1}</i>	2,117	849.74	1,068.92	118.00	433.00	1,202.00
<i>Unique Users_t</i>	1,685	3,812.21	11,954.56	59.00	320.00	1,983.00
<i>New Users_t</i>	1,685	2,346.44	7,367.64	22.00	150.00	1,179.00
<i>Returning Users_t</i>	1,685	904.50	2,634.35	9.00	85.00	484.00
<i>Resurrected Users_t</i>	1,685	412.85	1,295.67	4.00	29.00	199.00
Control Variables						
<i>Audit_{t-1}</i>	2,117	0.9306	0.2543	1.0000	1.0000	1.0000
<i>ETH Momentum_{t-1}</i>	2,117	0.4540	0.6656	0.0213	0.2888	0.8189
<i>Age_{t-1}</i>	2,117	2.0265	1.8837	1.0000	1.0000	3.0000
<i>Token Dummy_{t-1}</i>	2,117	0.6741	0.4688	0.0000	1.0000	1.0000
<i>Token Returns_t</i>	2,117	0.0832	0.6187	-0.1777	-0.0262	0.1058
<i>Token Volatility_{t-1}</i>	2,117	1.3627	0.8092	0.9996	1.1905	1.4509
<i>Market Cap_{t-1}</i>	2,117	187.48	467.70	16.30	33.41	108.35
<i>Token Turnover_{t-1}</i>	2,117	2.2442	16.7748	0.0023	0.0070	0.0257
Cross-sectional Variables						
<i>No. of Topics_{t-1}</i>	2,117	2.6330	1.2835	2.0000	3.0000	4.0000
<i>Content Difference_{t-1}</i>	2,117	0.7982	0.1698	0.8038	0.8746	0.9044
<i>Lines of TVL Code_{t-1}</i>	1,672	107.96	97.73	39.00	76.00	148.00
<i>Lines of Contract Code_{t-1}</i>	1,694	4,265.74	6,816.15	1,108.00	2,424.13	4,629.79
Other Dependent Variables						
<i>Large Wallets Inflow_{t-1}</i>	1,015	0.1130	1.5491	0.0069	0.0160	0.0350
<i>Large Wallets Outflow_{t-1}</i>	1,015	0.0356	0.1089	0.0049	0.0126	0.0298
<i>% Long-Horizon Wallets_{t-1}</i>	592	0.4072	0.3041	0.1184	0.3898	0.6672
<i>% Mid-Horizon Wallets_{t-1}</i>	592	0.5200	0.2791	0.2856	0.5192	0.7817

TABLE 2 – Continued

Panel B: Pairwise Correlation												
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) $Ln(TVL)_t$	1.0000											
(2) $Ln(Unique\ Users)_t$	0.6920	1.0000										
(3) $Ln(Blockchain\ Tweets)_{t-1}$	0.2783	0.3827	1.0000									
(4) $Ln(Total\ Words)_{t-1}$	0.2454	0.3184	0.7384	1.0000								
(5) $Audit_{t-1}$	0.2145	0.1155	0.1753	0.1363	1.0000							
(6) $ETH\ Momentum_{t-1}$	0.0909	0.1159	0.1255	0.0851	0.1159	1.0000						
(7) Age_{t-1}	0.0976	0.0558	0.0517	0.0477	0.1841	-0.0350	1.0000					
(8) $Token\ Dummy_{t-1}$	0.2523	0.1612	0.1425	0.1384	0.4694	0.0804	0.2911	1.0000				
(9) $Token\ Returns_t$	0.0210	0.0834	0.0581	0.0464	0.0397	0.0660	0.0207	0.0985	1.0000			
(10) $Token\ Volatility_{t-1}$	0.0066	0.0512	0.0945	0.0480	0.0315	0.1650	-0.1001	0.1203	0.0569	1.0000		
(11) $Market\ Cap_{t-1}$	0.4634	0.4237	0.1202	0.1192	0.1548	0.0777	0.2415	0.2552	-0.0377	-0.0607	1.0000	
(12) $Token\ Turnover_{t-1}$	0.1033	0.0575	0.0541	0.0434	0.0615	0.0410	-0.0570	0.1016	-0.0090	0.0167	0.1780	1.0000

TABLE 3

Impact of Blockchain-Related Tweets on TVL

This table reports coefficient estimates from OLS regressions of TVL on the number of blockchain-related tweets. Column 1 presents the results without fixed effects. Column 2 includes fixed effects. Column 3 additionally controls for market-related variables. Column 4 uses an alternative measure of disclosure, which is the total number of words used in blockchain-related tweets posted by DeFi platforms ($\ln(\text{Total Words})$). The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) $\ln(TVL)_t$	(2) $\ln(TVL)_t$	(3) $\ln(TVL)_t$	(4) $\ln(TVL)_t$
$\ln(\text{Blockchain Tweets})_{t-1}$	0.6234*** (4.7026)	0.4788*** (4.9306)	0.4416*** (4.5225)	
$\ln(\text{Total Words})_{t-1}$				0.1831*** (3.6419)
Audit_{t-1}	2.3996*** (5.3306)	0.9061*** (2.7171)	0.8012** (2.3530)	0.8027** (2.3047)
$\text{ETH Momentum}_{t-1}$	0.3367*** (2.9432)	0.5639*** (4.8176)	0.5274*** (4.6288)	0.5747*** (4.9854)
Age_{t-1}	0.1044 (0.6813)	-0.2055 (-0.2617)	-0.2403 (-0.3166)	-0.1841 (-0.2360)
Token Dummy_{t-1}			0.5163* (1.8161)	0.5262* (1.7822)
Token Returns_t			-0.0529 (-0.9742)	-0.0416 (-0.7568)
$\text{Token Volatility}_{t-1}$			0.0749 (1.2124)	0.1203** (1.9893)
Market Cap_{t-1}			0.0002 (0.7380)	0.0002 (0.5628)
$\text{Token Turnover}_{t-1}$			0.0106*** (3.4469)	0.0115*** (4.0378)
Platform, Year & Month FE	N	Y	Y	Y
Observations	2,117	2,117	2,117	2,117
Adjusted R-squared	0.1282	0.7970	0.8013	0.7958

TABLE 4

Information Content of Blockchain-Related Tweets

This table presents results using the textual characteristics of tweets to measure information content. Cross-sectional tests are conducted by partitioning on the median of *No. of Topics* (columns 1 and 2) and *Content Difference* (columns 3 and 4). The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) <i>Ln(TVL)_t</i> High <i>No. of</i> <i>Topics</i>	(2) <i>Ln(TVL)_t</i> Low <i>No. of</i> <i>Topics</i>	(3) <i>Ln(TVL)_t</i> High <i>Content</i> <i>Difference</i>	(4) <i>Ln(TVL)_t</i> Low <i>Content</i> <i>Difference</i>
<i>Ln(Blockchain Tweets)_{t-1}</i>	0.5349*** (4.5093)	0.2454** (2.5635)	0.6429*** (4.4546)	0.2652*** (2.6337)
<i>Audit_{t-1}</i>	0.5188 (1.4544)	1.2478* (1.9790)	0.2711 (0.8013)	1.2423** (2.5519)
<i>ETH Momentum_{t-1}</i>	0.4464*** (3.5198)	0.6113*** (3.6956)	0.4820*** (3.1099)	0.5890*** (3.7485)
<i>Age_{t-1}</i>	-0.4082* (-1.7746)	-0.4153 (-0.5124)	-0.4112* (-1.9401)	-0.1832 (-0.2079)
<i>Token Dummy_{t-1}</i>	0.4523 (1.4283)	0.1166 (0.2959)	0.4121 (1.4459)	0.3077 (0.7286)
<i>Token Returns_t</i>	-0.0571 (-1.0337)	-0.0002 (-0.0018)	-0.0673 (-0.9371)	-0.0088 (-0.1020)
<i>Token Volatility_{t-1}</i>	0.1695** (2.3698)	-0.0492 (-0.7547)	0.1167 (1.2859)	0.0225 (0.3292)
<i>Market Cap_{t-1}</i>	0.0004* (1.6716)	0.0002 (0.6802)	0.0004** (2.4232)	0.0003 (0.8330)
<i>Token Turnover_{t-1}</i>	0.0114*** (5.1730)	0.0201*** (4.3261)	0.0139*** (6.8587)	0.0132*** (3.1947)
Coef. Diff (p-value)	0.02**		0.02**	
Platform, Year & Month FE	Y	Y	Y	Y
Observations	1,252	850	1,057	1,052
Adjusted R-squared	0.8248	0.7842	0.8226	0.7889

TABLE 5

Information Processing Costs

This table presents cross-sectional results when partitioning the main sample on information processing costs. Columns 1 to 4 use two variables: lines of code that DeFi Pulse uses to calculate TVL (columns 1 and 2) and lines of smart contract code that the DeFi platform uses (columns 3 and 4). The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) $Ln(TVL)_t$ High Lines of TVL Code	(2) $Ln(TVL)_t$ Low Lines of TVL Code	(3) $Ln(TVL)_t$ High Lines of Contract Code	(4) $Ln(TVL)_t$ Low Lines of Contract Code
$Ln(Blockchain\ Tweets)_{t-1}$	0.7516*** (5.5520)	0.2556* (1.8267)	0.4352*** (4.3097)	0.1218 (1.2416)
$Audit_{t-1}$	1.6833*** (4.4286)	0.8692 (1.4434)	0.6717 (1.3597)	-0.1411 (-0.3834)
$ETH\ Momentum_{t-1}$	0.3175** (2.1253)	0.5513*** (3.7937)	0.4210** (2.6411)	0.4888*** (4.0226)
Age_{t-1}	1.0801 (1.1831)	-0.9716*** (-5.3268)	-1.2901*** (-6.2397)	9.2110*** (13.6656)
$Token\ Dummy_{t-1}$	1.0041** (2.2168)	-0.0398 (-0.1176)	0.3113 (0.9183)	0.2788 (0.8250)
$Token\ Returns_t$	-0.0132 (-0.1615)	-0.1329 (-1.5280)	0.0301 (0.5248)	-0.0025 (-0.0294)
$Token\ Volatility_{t-1}$	-0.1076 (-1.2621)	0.0382 (0.4529)	0.0373 (0.5532)	0.1648** (2.0023)
$Market\ Cap_{t-1}$	0.0007** (2.6591)	0.0001 (0.3235)	0.0008*** (2.7045)	0.0001 (0.2098)
$Token\ Turnover_{t-1}$	0.0061** (2.2624)	0.0947 (1.4827)	0.0092*** (4.7301)	0.0442 (1.6647)
Coef. Diff (p-value)	0.01**		0.03**	
Platform, Year & Month FE	Y	Y	Y	Y
Observations	815	857	882	803
Adjusted R-squared	0.8491	0.7919	0.8370	0.8728

TABLE 6

Impact of Blockchain-Related Tweets on User Base

This table reports coefficient estimates from OLS regressions of the number of platform users on blockchain-related tweets. Column 1 presents the results without fixed effects. Column 2 includes fixed effects. Columns 3–5 show the breakdown by user type. The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) <i>Ln(Unique Users)_t</i>	(2) <i>Ln(Unique Users)_t</i>	(3) <i>Ln(New Users)_t</i>	(4) <i>Ln(Returning Users)_t</i>	(5) <i>Ln(Resurrected Users)_t</i>
<i>Ln(Blockchain Tweets)_{t-1}</i>	0.6947*** (5.0580)	0.2241** (2.0402)	0.2232** (1.9938)	0.3310*** (2.9477)	0.0506 (0.5927)
<i>Audit_{t-1}</i>	0.9440** (2.0657)	0.8172** (2.1313)	0.5110 (1.2440)	1.4005*** (3.9309)	1.5638*** (4.6550)
<i>ETH Momentum_{t-1}</i>	0.2962*** (2.8924)	0.6114*** (5.0803)	0.7104*** (5.2596)	0.4242*** (4.4232)	0.0176 (0.1870)
<i>Age_{t-1}</i>	0.0489 (0.5034)	-0.8641** (-2.5033)	-1.0933*** (-3.0095)	-0.2402 (-0.7505)	-0.4036** (-2.4654)
<i>Token Dummy_{t-1}</i>		0.2465 (0.7831)	0.0161 (0.0458)	0.9095*** (2.8150)	1.1586*** (3.9766)
<i>Token Returns_t</i>		0.2288*** (4.2760)	0.2960*** (5.2350)	0.0723 (1.0326)	0.0868 (1.2580)
<i>Token Volatility_{t-1}</i>		0.1179 (1.4488)	0.1224 (1.6034)	0.1795** (2.0088)	-0.2142*** (-3.3469)
<i>Market Cap_{t-1}</i>		0.0005* (1.8062)	0.0006** (2.2574)	0.0004* (1.7077)	0.0004* (1.6682)
<i>Token Turnover_{t-1}</i>		0.0070** (2.0380)	0.0027 (0.3696)	0.0062* (1.7971)	0.0113*** (5.4319)
Platform, Year & Month FE	N	Y	Y	Y	Y
Observations	1,685	1,685	1,685	1,685	1,685
Adjusted R-squared	0.1639	0.7430	0.7207	0.7512	0.7830

TABLE 7

TVL and Voting Power Concentration

This table reports the coefficient estimates from OLS regressions of TVL user concentration and voting power concentration on blockchain-related tweets. Panel A columns 1 and 2 present the results from partitioning the sample on *TVL Concentration*, and column 3 regresses *TVL Concentration* on blockchain-related tweets. Panel B presents the results of regressing voting participation rate on blockchain-related tweets (column 1) and governance-related tweets (column 2). The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

Panel A: TVL Concentration			
	(1) <i>Ln(TVL)_t</i> High TVL <i>Concentration_t</i>	(2) <i>Ln(TVL)_t</i> Low TVL <i>Concentration_t</i>	(3) <i>TVL</i> <i>Concentration_t</i>
<i>Ln(Blockchain Tweets)_{t-1}</i>	0.0592 (0.6413)	0.4374*** (4.7896)	0.1619 (1.5301)
<i>Audit_{t-1}</i>	-0.1073 (-0.2968)	1.1099*** (2.8348)	0.3783 (0.9843)
<i>ETH Momentum_{t-1}</i>	0.0907 (0.7092)	0.5733*** (3.9897)	-0.0121 (-0.1158)
<i>Age_{t-1}</i>	0.9891* (1.7153)	-1.3364*** (-5.2001)	-0.3130 (-0.5546)
<i>Token Dummy_{t-1}</i>	0.4710 (1.3116)	0.2887 (0.7837)	0.3100 (1.0690)
<i>Token Returns_t</i>	-0.0768 (-1.5617)	0.1093* (1.6745)	-0.1657*** (-3.2530)
<i>Token Volatility_{t-1}</i>	0.0607 (1.0534)	0.1883** (2.6206)	-0.0296 (-0.5677)
<i>Market Cap_{t-1}</i>	-0.0002 (-0.8982)	0.0004 (1.4476)	-0.0002 (-0.7692)
<i>Token Turnover_{t-1}</i>	0.0213*** (3.8674)	0.0091 (1.5103)	0.0018 (0.4525)
Coef. Diff (p-value)	<0.01***		
Platform, Year & Month FE	Y	Y	Y
Observations	806	755	1,576
Adjusted R-squared	0.8953	0.7918	0.6619

TABLE 7 – Continued

Panel B: Voting Participation		
	(1) <i>VotePart_t</i>	(2) <i>VotePart_t</i>
<i>Ln(Blockchain Tweets)_{t-1}</i>	0.0237** (2.3437)	
<i>Ln(Governance Tweets)_{t-1}</i>		0.0145* (1.7926)
<i>Audit_{t-1}</i>	0.0894*** (3.2885)	0.0765*** (3.3828)
<i>ETH Momentum_{t-1}</i>	0.0088 (0.5763)	0.0088 (0.5853)
<i>Age_{t-1}</i>	0.0820* (1.9134)	0.0758* (1.7530)
<i>Token Dummy_{t-1}</i>	-0.0375 (-1.0375)	-0.0360 (-0.9166)
<i>Token Returns_t</i>	0.0104 (1.4437)	0.0080 (1.0812)
<i>Token Volatility_{t-1}</i>	-0.0011 (-0.1940)	-0.0007 (-0.1183)
<i>Market Cap_{t-1}</i>	-0.0000** (-2.1048)	-0.0000** (-2.1488)
<i>Token Turnover_{t-1}</i>	0.0004 (1.4424)	0.0004 (1.3430)
Platform, Year & Month FE	Y	Y
Observations	405	405
Adjusted R-squared	0.3753	0.3572

TABLE 8

Strategic Tweeting

This table reports coefficient estimates from OLS regressions of the number of blockchain-related tweets on different user base variables. Column 1 presents the results using large wallet outflows, and column 2 uses large wallet inflows. Column 3 presents the results using the percentage of mid-horizon wallets, and column 4 uses the percentage of long-horizon wallets. The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) <i>Ln(Blockchain Tweets)_t</i>	(2) <i>Ln(Blockchain Tweets)_t</i>	(3) <i>Ln(Blockchain Tweets)_t</i>	(4) <i>Ln(Blockchain Tweets)_t</i>
<i>Large Wallets Outflow_{t-1}</i>	0.4608** (2.3551)			
<i>Large Wallets Inflow_{t-1}</i>		-0.0133* (-1.6997)		
<i>% Mid-Horizon Wallets_{t-1}</i>			0.6297* (1.9759)	
<i>% Long-Horizon Wallets_{t-1}</i>				-0.6102* (-1.8559)
<i>Audit_{t-1}</i>	-0.0062 (-0.0346)	-0.0625 (-0.3375)	0.2691 (1.0585)	0.2181 (0.8647)
<i>ETH Momentum_{t-1}</i>	0.2277*** (4.2926)	0.2376*** (4.4399)	0.2752*** (2.8670)	0.2459** (2.5537)
<i>Age_{t-1}</i>	-0.2097 (-1.5172)	-0.2110 (-1.4774)	0.4194** (2.2825)	0.4545** (2.5074)
<i>Token Dummy_{t-1}</i>	0.1626 (1.0398)	0.0552 (0.3666)	0.2511 (0.9549)	0.2459 (0.9713)
<i>Token Returns_t</i>	0.0272 (0.8371)	0.0251 (0.7615)	0.0061 (0.1101)	0.0006 (0.0114)
<i>Token Volatility_{t-1}</i>	0.0515 (1.2438)	0.0747* (1.8984)	0.0211 (0.4485)	0.0076 (0.1652)
<i>Market Cap_{t-1}</i>	-0.0003** (-2.0608)	-0.0003** (-2.0584)	-0.0005** (-2.3020)	-0.0005** (-2.4086)
<i>Token Turnover_{t-1}</i>	0.0004 (0.1266)	0.0005 (0.1509)	0.0019 (0.4063)	0.0024 (0.5167)
<i>InstOwn_{t-1}</i>	-0.1275 (-0.1782)	0.0292 (0.0404)	1.3530 (0.8675)	1.3161 (0.8738)
Platform, Year & Month FE	Y	Y	Y	Y
Observations	1,015	1,015	592	592
Adjusted R-squared	0.6950	0.6941	0.6650	0.6649

TABLE 9

Contagion Risk from CEX Hacks

This table reports coefficient estimates from OLS regressions of platform users and TVL on blockchain-related tweets and *CEX Hack*. Column 1 presents the results for unique number of users. Columns 2 and 3 show the breakdown by user type. Column 4 presents the results for TVL. The t-statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

	(1) <i>Ln(Unique Users)_t</i>	(2) <i>Ln(Returning Users)_t</i>	(3) <i>Ln(New Users)_t</i>	(4) <i>Ln(TVL)_t</i>
<i>Ln(Blockchain Tweets)_{t-1}</i>	0.5766*** (4.7397)	0.6003*** (4.7564)	0.5824*** (4.6586)	0.5691*** (4.8028)
<i>CEX Hack_{t-1}</i>	-0.3268** (-2.0953)	-0.1858 (-1.1222)	-0.3888** (-2.2622)	-0.2336 (-1.3430)
<i>CEX Hack_{t-1} × Ln(Blockchain Tweets)_{t-1}</i>	0.1059* (1.7936)	0.1290** (2.1825)	0.1120* (1.7442)	0.1189* (1.8072)
<i>Audit_{t-1}</i>	0.3691 (0.8615)	1.0211** (2.2329)	0.2965 (0.6602)	1.3191*** (2.7543)
<i>ETH Momentum_{t-1}</i>	0.2450** (2.1740)	0.0955 (0.8522)	0.3250** (2.6015)	0.2800** (2.4133)
<i>Age_{t-1}</i>	-0.0310 (-0.3285)	-0.1046 (-1.0178)	-0.0026 (-0.0260)	-0.0247 (-0.2509)
<i>Token Dummy_{t-1}</i>	0.3430 (1.1365)	0.6951** (2.1912)	0.1528 (0.4838)	0.7399* (1.7537)
<i>Token Returns_t</i>	0.2730*** (3.8368)	0.1165 (1.4201)	0.3456*** (4.5105)	0.1001 (1.1581)
<i>Token Volatility_{t-1}</i>	0.0125 (0.1296)	0.0356 (0.3455)	0.0340 (0.3356)	-0.0778 (-0.8315)
<i>Market Cap_{t-1}</i>	0.0019*** (5.7565)	0.0019*** (5.6907)	0.0021*** (5.9520)	0.0024*** (5.8514)
<i>Token Turnover_{t-1}</i>	-0.0042 (-0.5119)	-0.0029 (-0.3639)	-0.0069 (-0.7007)	0.0012 (0.1698)
Category, Year & Month FE	Y	Y	Y	Y
Observations	1,576	1,576	1,576	2,117
Adjusted R-squared	0.3553	0.3663	0.3532	0.4185

TABLE 10

Robustness Tests

Panel A presents an estimate of the value of Delta (δ), the relative degree of selection on observed and unobserved control variables for which the independent variable of interest ($\ln(\text{Blockchain Tweets})$) would be 0, following Oster (2019). The coefficient on $\ln(\text{Blockchain Tweets})$ and R^2 are shown from the estimation of the main specification (Table 3 column 3), with $(\beta_{\text{Controlled}}, R^2_{\text{Controlled}})$ and without $(\beta_{\text{Uncontrolled}}, R^2_{\text{Uncontrolled}})$ control variables. Following Oster (2019), I set R_{max} (R^2 from a hypothetical regression of the outcome on treatment and both observed and unobserved control variables) equal to 1.3 (and 1.5 for robustness) multiplied by R^2 of the regression that includes all control variables (i.e., the controlled regression). Panel B presents results from the 2SLS regression at the platform-day level, with *Twitter Outage* as the instrument. For Panel B, t -statistics are based on standard errors clustered at the platform level. Asterisks (***, **, and *) indicate statistical significance at the 1, 5, and 10 percent levels, respectively, based on two-tailed tests.

Panel A: Oster (2019) Analysis		
Independent Variable = $\ln(\text{Blockchain Tweets})_{t-1}$	Dependent Variable = $\ln(\text{TVL})_t$	
	(1) $R_{\text{max}} = 1.3 \times R^2$	(2) $R_{\text{max}} = 1.5 \times R^2$
$\beta_{\text{Uncontrolled}}$	0.495	0.495
$R^2_{\text{Uncontrolled}}$	0.043	0.043
$\beta_{\text{Controlled}}$	0.441	0.441
$R^2_{\text{Controlled}}$	0.510	0.510
δ	12.998	8.035

TABLE 10 - Continued

Panel B: IV Approach		
	(1) <i>Ln(Blockchain Tweets)_t</i>	(2) <i>Ln(TVL)_t</i>
<i>Twitter Outage_{t-1,t}</i>	-0.0260*** (-3.1766)	
<i>Predicted Ln(Blockchain Tweets)_{t-1}</i>		1.9212** (2.0742)
<i>Audit_{t-1}</i>	-0.2175*** (-2.6540)	1.2699*** (2.6684)
<i>ETH Momentum_{t-1}</i>	0.0418** (2.0538)	-0.1409 (-1.4203)
<i>Age_{t-1}</i>	0.0083 (0.3201)	-1.0412** (-2.5363)
<i>Token Dummy_{t-1}</i>	0.0202 (0.3178)	-0.0554 (-0.3060)
<i>Token Volatility_{t-1}</i>	0.0267*** (2.9572)	-0.1673*** (-3.1617)
<i>Market Cap_{t-1}</i>	-0.0001* (-1.9745)	0.0004 (1.3812)
<i>Token Turnover_{t-1}</i>	0.0068 (0.1295)	0.3874* (1.7708)
Platform & Year FE	Y	Y
Observations	41,073	41,073
Adjusted R-squared	0.2994	0.8540